

PAGO PAGO, American Samoa

WMO: 917650

Lat: 14.331S

Lon: 170.714W

Elev: 4

StdP: 101.28

Time Zone: -11.00 (W11)

Period: 94-19

WBAN: 61705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.2	23.8	18.7	13.5	26.2	19.8	14.5	26.3	12.1	26.3	11.4	26.6	2.2	300	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	5.4	32.6	27.3	32.1	27.1	31.7	27.0	28.6	30.9	28.2	30.6	27.9	30.3	5.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.1	30.1	27.6	23.5	29.8	27.2	22.9	29.6	92.5	31.0	90.7	30.7	89.2	30.4	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.9	10.1	9.3	DB	21.7	33.9	1.1	1.4	20.9	34.9	20.3	35.7	19.7	36.5	18.9	37.4
			WB	19.0	29.2	1.1	0.8	18.2	29.8	17.5	30.2	16.9	30.7	16.1	31.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	28.4	28.5	28.5	28.4	27.9	27.6	27.2	27.2	27.6	27.8	28.1	28.3	(6)
	DBStd	1.14	1.06	1.13	1.00	1.00	1.04	0.96	1.05	1.01	1.02	1.05	1.06	1.02	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6551	571	518	574	551	554	527	532	532	529	552	544	568	(10)
	CDD18.3	3509	312	284	316	301	296	277	274	273	279	294	294	310	(11)
	CDH23.3	39620	3721	3405	3790	3573	3306	2988	2783	2752	2978	3240	3423	3661	(12)
	CDH26.7	12884	1404	1320	1474	1342	1047	793	615	611	797	967	1183	1331	(13)
(14) Wind	WSAvg	4.9	4.0	4.0	3.6	3.9	4.7	5.9	6.0	5.9	6.1	5.3	4.6	4.3	(14)
(15) Precipitation	PrecAvg	2990	338	315	289	291	261	164	152	168	186	268	277	341	(15)
	PrecMax	3889	710	830	651	720	739	320	365	470	642	739	652	685	(16)
	PrecMin	1519	118	51	102	19	41	67	18	9	25	53	82	57	(17)
	PrecStd	518	153	156	137	168	159	73	86	115	137	145	127	153	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.3	33.1	32.8	31.9	31.1	30.9	30.9	31.6	32.1	32.8	32.8	(19)
		MCWB	27.2	27.7	27.4	27.4	26.8	26.6	26.2	26.3	26.5	27.1	27.3	27.3	(20)
	2%	DB	32.1	32.2	32.2	31.9	31.1	30.5	30.2	30.2	30.8	31.2	31.8	31.9	(21)
		MCWB	27.1	27.2	27.3	27.2	26.6	26.3	25.9	25.9	26.4	26.9	27.1	27.0	(22)
	5%	DB	31.6	31.7	31.7	31.3	30.6	30.0	29.7	29.8	30.2	30.6	31.1	31.3	(23)
		MCWB	27.0	27.1	27.2	27.0	26.5	26.2	25.8	25.8	26.1	26.6	27.0	26.9	(24)
	10%	DB	31.0	31.1	31.1	30.9	30.1	29.5	29.2	29.2	29.7	30.0	30.6	30.8	(25)
		MCWB	26.9	26.9	27.0	26.9	26.3	26.0	25.5	25.5	25.9	26.3	26.8	26.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	28.8	29.1	28.7	28.2	27.7	27.7	27.9	28.1	28.5	28.6	28.7	(27)
		MCDB	31.1	31.5	31.2	30.8	30.3	29.6	29.2	29.5	30.2	30.6	30.9	30.8	(28)
	2%	WB	28.1	28.2	28.2	28.2	27.6	27.1	26.9	26.9	27.3	27.7	27.9	28.0	(29)
		MCDB	30.7	30.8	30.8	30.4	29.8	29.3	28.8	28.7	29.6	29.9	30.4	30.4	(30)
	5%	WB	27.7	27.8	27.7	27.7	27.2	26.7	26.5	26.5	26.8	27.2	27.5	27.6	(31)
		MCDB	30.4	30.4	30.5	30.2	29.5	29.0	28.6	28.5	29.1	29.5	30.1	30.1	(32)
	10%	WB	27.2	27.3	27.3	27.3	26.8	26.4	26.1	26.1	26.4	26.7	27.1	27.2	(33)
		MCDB	30.1	30.2	30.3	29.9	29.1	28.7	28.3	28.3	28.7	29.1	29.8	29.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.1	5.4	5.0	4.4	3.8	3.9	4.0	4.3	4.4	4.7	4.9	(35)
		MCDBR	5.8	5.8	5.8	5.5	5.0	4.4	4.5	4.7	4.8	5.0	5.3	5.5	(36)
	5% WB	MCWBR	2.8	2.6	2.8	2.7	2.6	2.4	2.5	2.6	2.6	2.6	2.7	2.7	(37)
		MCWBR	5.1	5.1	5.3	5.0	4.7	4.2	4.3	4.4	4.6	4.8	5.0	5.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.387	0.378	0.374	0.382	0.389	0.392	0.392	0.397	0.399	0.386	0.383	0.385	(40)
		taud	2.633	2.670	2.686	2.656	2.618	2.590	2.584	2.562	2.562	2.609	2.637	2.635	(41)
	Ebn at Noon		956	957	941	901	861	841	850	874	904	938	954	957	(42)
		Edh at Noon	101	97	93	90	89	88	91	98	103	102	100	101	(43)
(44) All-Sky Solar Radiation	RadAvg		5.91	5.88	5.86	5.17	4.55	4.35	4.50	5.21	5.85	6.10	6.01	5.95	(44)
	RadStd		0.59	0.52	0.44	0.43	0.32	0.21	0.18	0.24	0.34	0.33	0.29	0.33	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page